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How Much Local Data Is Enough?

An ontology-grounded neurosymbolic research agenda
for transferability in forecasting

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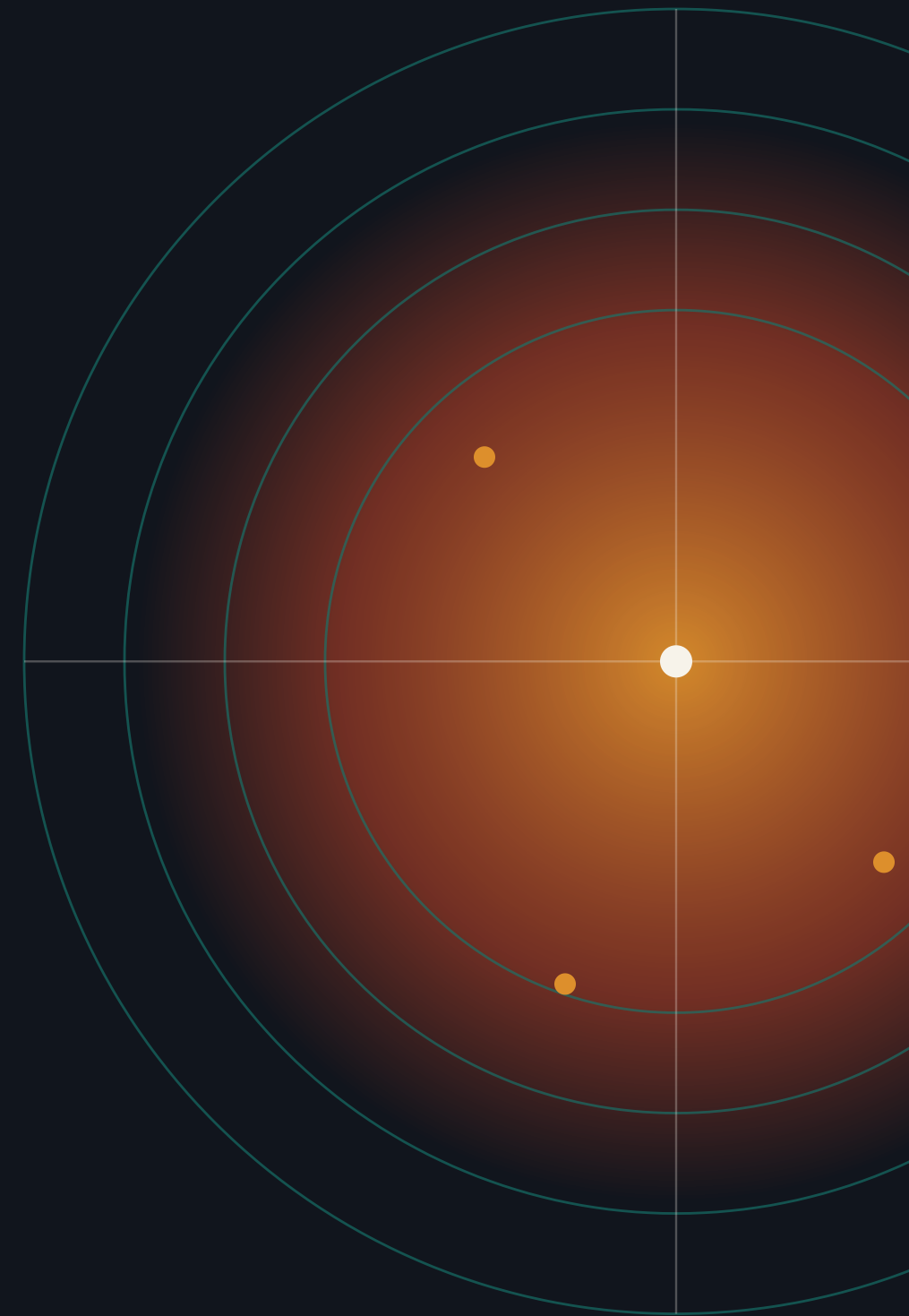
NEUROSymbOLIC AI

WELL-FOUNDED ONTOLOGY

TRANSFER LEARNING

SOLAR IRRADIATION

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Brazil measures the sun in a *few places*, and almost nowhere else.

- Renewable-energy planning needs reliable solar-irradiation forecasts, and good forecasts need data.
- A handful of well-instrumented sites have years of clean records.
- Vast regions have almost nothing — a continent-sized country with a sparse and uneven measurement network.

The requirement is rarely met — yet the forecasts are needed anyway.

Train where data is plentiful. Leverage where it is not.

That is **transfer learning** [Pan & Yang 2010] — and it is not epistemically neutral.

*Leveraging presupposes that the target is, in some relevant respect,
the same kind of place as the source.*

“The same kind of place” is *not a statistical primitive.*

It is a judgment about which properties of a site are **essential** to its solar behavior and which are **incidental**.

TWO STATIONS MAY

share a latitude
but differ in cloud regime

TWO STATIONS MAY

share a biome
but differ in altitude

TWO STATIONS MAY

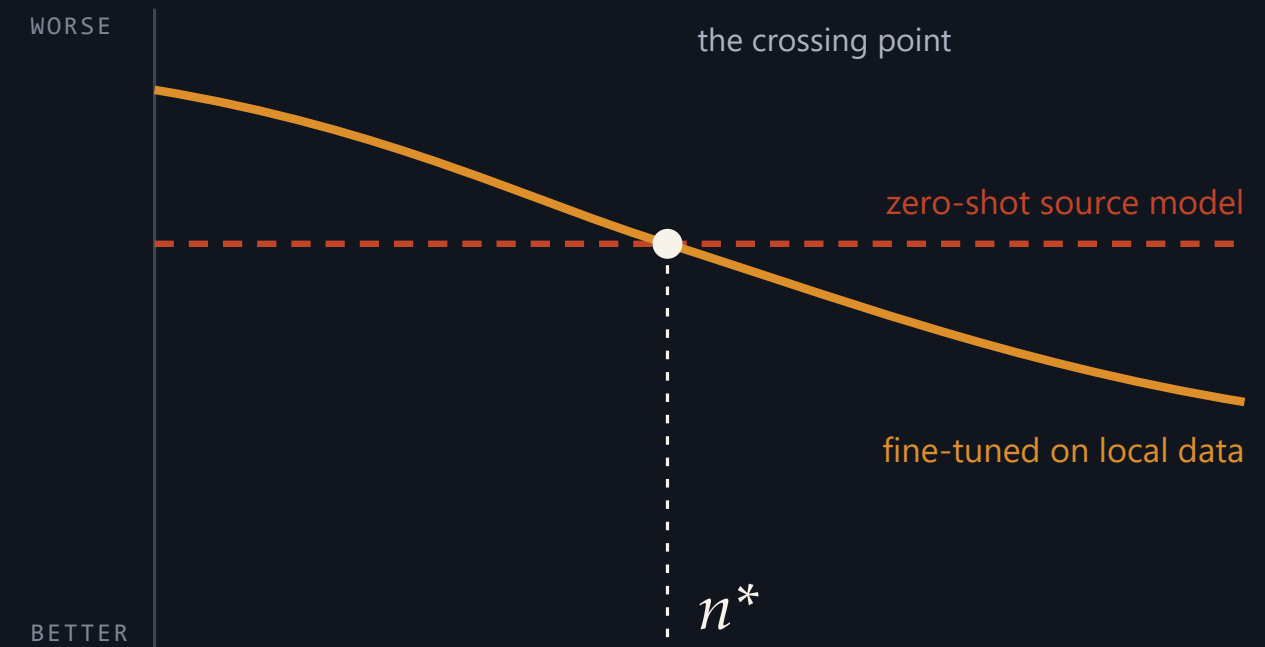
sit far apart
yet behave identically

When source and target diverge along the dimensions that matter, transfer **degrades performance rather than improving it** — negative transfer. [Pan & Yang 2010; Weiss et al. 2016]

How much local data is *enough*?

Given a source model and a target site: how much local data must be collected before **fine-tuning beats deploying the source model as-is?**

Call that quantity the site's **data budget, n^*** .

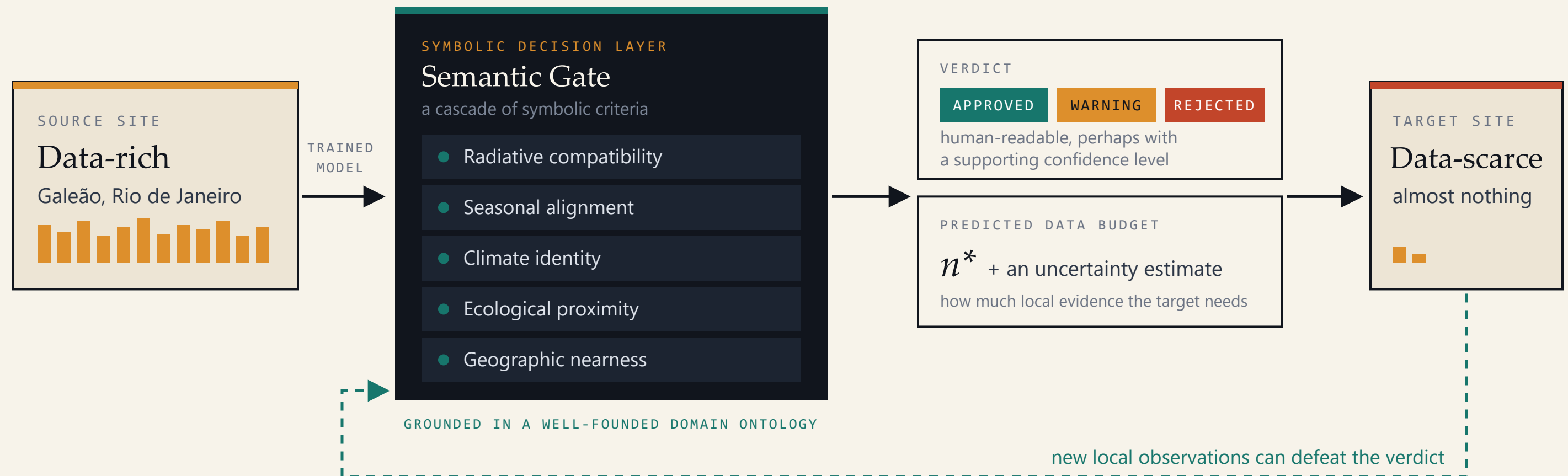


LOCAL OBSERVATIONS AT THE TARGET →
SCHEMATIC — DEFINITIONAL, NO MEASURED VALUES SHOWN

A theory of n^* would let practitioners estimate, **before investing in instrumentation**, how much data is likely to be required.

A *Neurosymbolic Semantic Gate* between the model and the place.

A symbolic decision layer, still under development, that reasons — transparently, in terms a domain scientist can read — about whether and how readily transfer is warranted.



Each criterion is a *distinct, testable hypothesis* about what licenses transfer.

CRITERION	NOTION OF “SAMENESS”	ONTOLOGICAL SOURCE
Radiative compatibility	Comparable top-of-atmosphere forcing	Astronomy / site geometry
Seasonal alignment	Matching shape of the annual cycle	Climatology
Climate identity	Same Köppen class	Köppen–Geiger classification
Ecological proximity	Near in biome / climate space	Whittaker biome scheme
Geographic nearness	Small great-circle distance	Spatial proximity

These are **distinct relations that often disagree**. The design keeps them separate so the study can ask: **which notion of sameness, if any, actually carries predictive weight?**

Two claims I intend to put *under test*, not to assume.

H1 — ORDERING

If ontological similarity orders source–target pairs, then sites judged **more similar** require fewer local observations to adapt — a measurably lower data budget n^* .

H2 — DISCRIMINATION

If the candidate notions of sameness are **not interchangeable**, some (categorical climate identity) will carry systematically different inductive value from others (graded behavioral proximity) — and the empirical record will differentiate them.

A standard of acceptance that has never been challenged is **indistinguishable from an assumption**.

The gate will be measured against a *ground truth* that already exists.

The reference budget n^* has been established across 20 Brazilian sites, using reanalysis weather data [MERRA-2].

1

SOURCE SITE
GALEÃO, RIO DE JANEIRO

19

TARGET SITES

11–2,100

KM SPAN, TROPICAL RIO
BASIN TO THE
EQUATORIAL NORTH

SOURCE

EQUATORIAL NORTH



11 km — nearest target



≈ 2,100 km — farthest target

Each criterion, and the gate's aggregate verdict, will then be assessed for its ability to **rank target sites by n^*** — controlling for target-site difficulty, so the effect is not an artifact of a site being inherently easier to forecast.

Three overlapping work packages, *not* a sequence of phases.

WP1 · M1-M3

Measure the budget

n^* measured for all 20 sites under a fixed protocol. Delivers the ground-truth reference the gate is evaluated against.

COMPLETE

WP2 · M2-M9

Formalize & build the gate

The well-founded ontology and the gate that operationalizes it: source and target as **relational roles**, the graded relations that mediate them, and the epistemic status of a verdict.

MOSTLY COMPLETE

WP3 · M7-M12

Analyze against measured budgets

Test Ordering and Discrimination. Results directly validate or falsify the central claims ahead of the write-up.

TESTING PHASE

*Transferability is a structured, investigable phenomenon — and formal ontology has a **concrete, testable role** in explaining it.*

Not decorative support for a machine-learning pipeline, but a **source of hypotheses a learning system can evaluate**. The same skeleton — a source model, a new context, a local adaptation budget, competing similarity relations — recurs in hydrology, air quality, agriculture and other renewables.

PUSHBACK WANTED ON

How I individuate and operationalize the similarity relations.

PUSHBACK WANTED ON

The epistemic status I assign to a gate's verdict.

PUSHBACK WANTED ON

How to frame the ontology–learning interface so it speaks to both communities.

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